DYNAMIC INPATIENT PROGNOSTIC MODELING: A SEMI-PARAMETRIC SURVIVAL ANALYSIS AND COUNTERFACTUAL DECISION-SUPPORT SYSTEM

By

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Dynamic Inpatient Prognostic Modeling leverages semi-parametric survival analysis to deliver real-time, time-varying risk trajectory forecasts for hospitalized patients.

The system integrates a counterfactual decision-support engine to simulate and evaluate "what-if" clinical interventions, addressing confounding and treatment selection bias.

Engineered as an interpretable clinical AI framework to evaluate treatment timing and optimize personalized bedside interventions.

🖇 **Model Design & Source Code:** https://github.com/rohit2205200

Deployed Model :https://rohitvpatil22052005.shinyapps.io/Survival/

ABSTRACT

Congestive heart failure remains a primary contributor to worldwide inpatient mortality and critical rehospitalizations. Conventional clinical triage paradigms predominantly depend on static categorical scorecards or binary classification estimators that artificially compress follow-up duration into crude 30-day binary endpoints, routinely introducing survivorship distortion when dealing with right-censored data.

In this investigation, we developed an individualized, dynamic prognostic survival engine operating on an inpatient clinical cohort ($N = 299$, 96 terminal failure events). Rather than imposing arbitrary parametric distributional families on time-to-event intervals, we implemented a multivariable Cox Proportional Hazards architecture optimized via Cox's partial likelihood. The model identifies left ventricular ejection fraction ($HR \approx 0.96$, 95% CI: 0.94--0.98, $p < 0.001$), serum creatinine ($HR \approx 1.30$, 95% CI: 1.15--1.48, $p < 0.001$), and chronological age ($HR \approx 1.05$, 95% CI: 1.02--1.07, $p < 0.001$) as primary independent prognostic drivers.

The stationary assumption of proportional hazards was verified across all individual parameters and the global metric using Grambsch--Therneau tests on scaled Schoenfeld residuals ($p > 0.05$). Reconstructing the non-parametric cumulative baseline hazard through Breslow's estimator, we mapped relative hazard ratios back to continuous survival probabilities $\hat{S}(t \mid X)$. Finally, the pipeline was deployed as a real-time reactive R Shiny web application, enabling bedside counterfactual "what-if" simulations, risk stratification, and patient-specific survival curve visualizations without software licensing overhead.

CHAPTER 1: INTRODUCTION & PROBLEM FORMULATION

1.1 Clinical Burden of Heart Failure

Cardiovascular pathologies, particularly congestive heart failure, represent an acute healthcare burden globally. Despite therapeutic improvements, in-hospital and post-discharge mortality remain elevated. A critical clinical challenge in intensive coronary units and emergency wards is accurate risk stratification: identifying which patients present elevated hazards of imminent decompensation versus those likely to remain stable over a 90- to 180-day trajectory.

1.2 The Flaws of Legacy Systems and Naive Machine Learning

Existing bedside triage tools (e.g., NYHA classifications, legacy point scorecards) present fundamental structural deficiencies:

1. **Coarse Discretization:** Continuous physiological biomarkers are reduced to arbitrary integer point buckets (e.g., Creatinine > 1.5 mg/dL $= 1$ point), eliminating physiological resolution and creating step-cliff artifacts at arbitrary thresholds.
2. **Binary Time Truncation:** Standard classification approaches (logistic regression, basic random forests) enforce a static binary label ($Y \in \{0,1\}$ at 30 days). A patient who experiences cardiac death on Day 2 is treated identically to one who fails on Day 29, while patients surviving past Day 30 contribute no longitudinal insight.
3. **The Censoring Trap:** Standard regression and machine learning architectures cannot natively accommodate right-censored observations (patients discharged, relocated, or surviving past the study cutoff). Deleting censored records introduces severe selection bias, whereas treating them as surviving indefinitely introduces survivorship bias.

1.3 Project Objectives

- Implement an auditable, regulatory-grade time-to-event pipeline respecting non-informative right-censoring mechanisms.
- Derive adjusted hazard ratios (HR) using multivariable Cox Proportional Hazards regression to quantify independent biomarker risk contributions.
- Formally validate model assumptions through Grambsch--Therneau tests on Schoenfeld residuals and quantify discrimination via Harrell's Concordance Index (C-index).
- Recover the continuous baseline survival function using the Breslow estimator to compute $\hat{S}(t \mid X_{\text{new}})$ dynamically.
- Build and deploy an open-access, reactive clinical decision-support application (R Shiny) for real-time counterfactual "what-if" prognostic evaluations.

1.4 Methodological Paradigm Shift

- Traditional cardiovascular risk assessment in routine clinical practice has long relied on blunt integer scorecards, static 30-day binary classification, or naive machine-learning models that mismanage right-censored data and collapse continuous biomarker gradients into coarse, arbitrary bins.
- By replacing categorical cut-offs with continuous partial-likelihood optimization and static nomograms with reactive Breslow trajectory recovery, this framework transforms clinical risk prediction from a coarse diagnostic checklist into a quantitative, patient-specific prognostic tool. It preserves the complete time-to-event continuum, fully accounts for censored clinical follow-up without survivorship bias, and delivers an auditable, real-time bedside engine capable of dynamic counterfactual simulation.

1.5 Unique Selling Proposition (USP) & Methodological Contribution

- The primary innovation of this investigation lies in resolving the critical translational bottleneck that separates theoretical biostatistics from real-time clinical bedside deployment. While semi-parametric survival models and partial likelihood estimation serve as regulatory benchmarks in global pharmaceutical trials, their clinical utility remains largely confined to retrospective analyses, static hazard ratio tables, and population-averaged survival plots. Concurrently, conventional emergency triage protocols continue to rely on coarse, point-based risk scorecards or naive machine learning classifiers that artificially compress follow-up duration into binary 30-day endpoints and inherently mismanage right-censored observations.

  This project bridges that divide by engineering an end-to-end, mathematically verified decision-support architecture. By recovering the continuous non-parametric cumulative baseline hazard via Breslow's estimator on the fly, the system maps instantaneous patient biomarkers directly into dynamic, individualized survival trajectories $\hat{S}(t \mid X_{\text{new}})$ accompanied by robust log-log confidence intervals. Furthermore, the reactive platform enables counterfactual "what-if" clinical simulations—allowing clinicians to quantitatively project the survival gains of therapeutic biomarker modulation (such as acute renal stabilization) prior to high-risk interventions—delivered through a zero-cost, open-access cloud interface that eliminates proprietary software overhead for resource-constrained healthcare settings.

1.6 Sectoral Significance and Practical Impact

- **Biostatistics & Statistical Programming**
  - Moves Beyond Static Reporting: Modernizes traditional biostatistics by transitioning from passive tables and frozen .Rmd reports into real-time, interactive micro-services.

    - Preserves Mathematical Integrity: Proves that complex asymptotic concepts—such as partial likelihood, Breslow baseline recovery, and Schoenfeld residual validation—can run dynamically without relying on black-box heuristics or sacrificing mathematical rigor.
    - End-to-End Pipeline Standard: Demonstrates a clean, reproducible workflow from raw clinical data cleaning (gtsummary, tidyverse) to diagnostic checks (cox.zph) and interactive user-facing dashboards (shiny).
- **Clinical Research Organizations (CROs) & Pharma Development**
    - Adaptive Trial Stratification: Enables clinical trialists to balance patient cohorts across baseline hazard tiers during randomization, reducing bias in multi-arm trials.
    - Synthetic Counterfactual Modeling: Allows principal investigators to conduct in-silico "what-if" simulations, evaluating how target reductions in specific adverse biomarkers would alter expected event-free survival before protocol finalization.
    - Regulatory-Ready Deliverables: Leverages proportional hazards modeling and survival distributions aligned with standard CDISC and FDA/ICH-E9 statistical guidelines.
- **Clinical Practice & Hospital Triage**
    - Bedside Decision Acceleration: Converts complex biomarker interactions into a single, intuitive survival curve $\hat{S}(t \mid X_{\text{new}})$ that can be evaluated during emergency ward rounds in seconds.
    - Cardiorenal Interaction Visibility: Clearly displays competing physiological trade-offs—such as the balance between improving cardiac pumping efficiency (EF) and aggravating renal stress (Creatinine).
    - Individualized Risk Trajectories: Replaces blunt 30-day binary scorecards with flexible continuous tracking (30, 90, 180, and 250-day outlooks) tailored to each patient.
- **Public Health & Healthcare Administration**
    - Zero Licensing Costs: Eliminates reliance on expensive proprietary enterprise informatics packages, making advanced prognostic tools available to district hospitals and primary health centers (PHCs).
    - Scalable Telemedicine Support: The lightweight browser architecture enables remote healthcare workers to enter basic lab values and receive risk-tier guidance for emergency tertiary transfers.
    - Transparent and Auditable Care: Provides a clear, non-proprietary audit trail for clinical governance, ensuring predictions are based on verifiable statistical models rather than hidden algorithms.

CHAPTER 2: THEORETICAL & MATHEMATICAL FOUNDATIONS

2.1 The Observation Space & Censoring Formulation

Let $T_i \in (0, \infty)$ denote the true failure time for subject i $(i = 1, \dots, n)$, and let $C_i \in (0, \infty)$ denote the corresponding right-censoring time. The observed follow-up interval $\tilde{T}_i$ and event indicator δ_i are defined as:

$$\tilde{T}_i = \min(T_i, C_i)$$

$$\delta_i = \mathbb{I}(T_i \leq C_i) = \begin{cases} 1 & \text{if failure is observed} \\ 0 & \text{if observation is right-censored} \end{cases}$$

The fundamental regularity condition is **Independent (Non-informative) Censoring**:

$$T_i \perp C_i \mid X_i$$

Conditional on baseline covariates X_i, the probability distribution of censoring time provides no information regarding the prospective hazard of failure.

2.2 Non-Parametric Survival: The Kaplan-Meier Estimator

Before covariate adjustment, univariable survival distributions are estimated using the product-limit estimator (Kaplan & Meier, 1958):

$$\hat{S}(t) = \prod_{j:t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right)$$

Where $t_{(1)} < t_{(2)} < \cdots < t_{(k)}$ represent distinct ordered failure times, d_j is the number of observed failure events at $t_{(j)}$, and $n_j = \sum_{m=1}^{n} \mathbb{I}\left(\tilde{T}_m \geq t_{(j)}\right)$ is the count of subjects at risk immediately prior to $t_{(j)}$. Two-group survival curve divergences are formally tested using the non-parametric Mantel--Haenszel Log-Rank test.

2.3 The Semi-Parametric Cox Proportional Hazards Model

To evaluate multiple continuous and discrete biomarkers simultaneously without imposing restrictive parametric assumptions (e.g., Weibull, Exponential, Log-Normal), the instantaneous hazard function $h(t \mid X_i)$ is formulated as (Cox, 1972):

$$h(t \mid X_i) = \lim_{\Delta t \to 0^+} \frac{P(t \leq T_i < t + \Delta t \mid T_i \geq t, X_i)}{\Delta t} = h_0(t) \exp(\boldsymbol{\beta}^T X_i)$$

- $h_0(t)$: Arbitrary, non-negative, unspecified baseline hazard function when $X = \mathbf{0}$.
- $\boldsymbol{\beta} = \left(\beta_1, \dots, \beta_p\right)^T$: Vector of log-hazard regression coefficients.
- $\exp(\boldsymbol{\beta}^T X_i)$: Relative risk multiplier (link function).

For two subjects with covariate profiles X_A and X_B, the hazard ratio (HR) is:

$$\frac{h(t \mid X_A)}{h(t \mid X_B)} = \frac{h_0(t) \exp(\boldsymbol{\beta}^T X_A)}{h_0(t) \exp(\boldsymbol{\beta}^T X_B)} = \exp\left(\boldsymbol{\beta}^T (X_A - X_B)\right)$$

The ratio is strictly time-invariant; the nuisance baseline hazard $h_0(t)$ cancels out completely.

2.4 Parameter Estimation via Cox Partial Likelihood

Because $h_0(t)$ is infinite-dimensional, classical Maximum Likelihood Estimation (MLE) is intractable. Parameter vector $\boldsymbol{\beta}$ is estimated by maximizing the **Cox Partial Likelihood**, conditioned exclusively upon the active risk set $\mathcal{R}(t_{(j)}) = \{m: \tilde{T}_m \geq t_{(j)}\}$:

$$L_p(\boldsymbol{\beta}) = \prod_{j=1}^{k} \frac{\exp(\boldsymbol{\beta}^T X_j)}{\sum_{m \in \mathcal{R}(t_{(j)})} \exp(\boldsymbol{\beta}^T X_m)}$$

Taking the natural logarithm yields:

$$\ell_p(\boldsymbol{\beta}) = \sum_{j=1}^{k} \left[\boldsymbol{\beta}^T X_j - \log\left(\sum_{m \in \mathcal{R}(t_{(j)})} \exp(\boldsymbol{\beta}^T X_m) \right) \right]$$

The score vector $U(\boldsymbol{\beta}) = \nabla \ell_p(\boldsymbol{\beta}) = \mathbf{0}$ is solved iteratively via the **Newton-Raphson algorithm**:

$$\boldsymbol{\beta}^{(m+1)} = \boldsymbol{\beta}^{(m)} + \left[\mathcal{I}(\boldsymbol{\beta}^{(m)})\right]^{-1} U(\boldsymbol{\beta}^{(m)})$$

Where $\mathcal{I}(\boldsymbol{\beta}) = -\nabla^2 \ell_p(\boldsymbol{\beta})$ is the empirical Fisher Information matrix, providing asymptotic covariance estimates $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \mathcal{I}^{-1}(\hat{\boldsymbol{\beta}})$.

2.5 Dynamic Survival Probability Recovery via the Breslow Estimator

While $\boldsymbol{\beta}$ provides relative hazard ratios, generating personalized absolute survival curves $\hat{S}(t \mid X_{\text{new}})$ requires recovering the underlying baseline hazard. We evaluate the cumulative baseline hazard $H_0(t) = \int_0^t h_0(u)\, du$ via the non-parametric **Breslow estimator**:

$$\hat{H}_0(t) = \sum_{j: t_{(j)} \leq t} \frac{d_j}{\sum_{m \in \mathcal{R}(t_{(j)})} \exp(\hat{\boldsymbol{\beta}}^T X_m)}$$

Connecting survival probability to cumulative hazard through $S(t) = \exp(-H(t))$, the individualized trajectory for profile X_{new} is computed as:

$$\hat{S}_{\text{new}}(t) = \exp\left(-\hat{H}_0(t) \exp(\hat{\boldsymbol{\beta}}^T X_{\text{new}})\right) = \left[\hat{S}_0(t)\right]^{\exp(\hat{\boldsymbol{\beta}}^T X_{\text{new}})}$$

Pointwise 95% confidence intervals are generated using the Delta Method on the log-log scale:

$$\mathrm{Var}\left(\log\left(-\log \hat{S}(t)\right)\right) = \frac{1}{\left(\log \hat{S}(t)\right)^2} \frac{\mathrm{Var}(\hat{S}(t))}{\left(\hat{S}(t)\right)^2}$$

This guarantees that transformed confidence bounds remain bounded within the valid probability range [0,1].

CHAPTER 3: DATA PREPROCESSING & EXPLORATORY PROFILING

3.1 Dataset Structure

The dataset comprises $N = 299$ heart failure patients admitted to specialized inpatient care, with clinical follow-up tracking survival over an observation period of up to 285 days. The dataset exhibits zero missing values across critical biomarkers. A total of 96 terminal failure events ($\delta = 1$, 32.1%) and 203 right-censored observations ($\delta = 0$, 67.9%) are recorded.

3.2 Clinical Biomarker Cohort Profile (Table 1)

Clinical Parameter	Overall Cohort (N=299)	Event / Deceased (N=96)	Censored / Survived (N=203)	Test Metric (p-value)
Age (Years)	60.8 ± 11.9	65.2 ± 13.2	58.8 ± 10.7	$t = 4.12$ ($p < 0.001$)
Sex (Male, %)	194 (64.9%)	62 (64.6%)	132 (65.0%)	$\chi^2 = 0.005$ ($p = 0.941$)
Ejection Fraction (%)	38.1 ± 11.8	33.5 ± 10.5	40.3 ± 11.7	$t = -4.95$ ($p < 0.001$)
Serum Creatinine (mg/dL)	1.39 ± 1.03	1.84 ± 1.47	1.18 ± 0.65	$W = 5821$ ($p < 0.001$)
Serum Sodium (mEq/L)	136.6 ± 4.4	135.4 ± 5.0	137.2 ± 4.0	$t = -3.07$ ($p = 0.002$)
Hypertension (Yes, %)	105 (35.1%)	40 (41.7%)	65 (32.0%)	$\chi^2 = 2.37$ ($p = 0.123$)

CHAPTER 4: MODELING, DIAGNOSTICS & VERIFICATION

4.1 Kaplan-Meier Stratification by Systolic Function

Unadjusted survival divergence was evaluated by stratifying patients by clinical left ventricular impairment (EF $< 30\%$ vs EF $\geq 30\%$).

- Patients presenting with severely depressed ejection fraction ($< 30\%$) exhibited early and sustained survival declines.
- The non-parametric Log-Rank test yielded $\chi^2 = 18.6$ ($p < 0.0001$), confirming significant survival disparity prior to multivariable adjustment.

4.2 Multivariable Cox Proportional Hazards Model Results

Covariate	Coef (β^)	SE(β^)	Hazard Ratio (exp(β^))	95% CI (Lower, Upper)	Wald z	p-value
Age	0.046	0.009	1.047	[1.028,1.066]	4.96	< 0.001
Sex (Male vs Female)	-0.187	0.222	0.829	[0.537,1.282]	-0.84	0.400
Ejection Fraction	-0.038	0.010	0.963	[0.943,0.982]	-3.71	< 0.001
Serum Creatinine	0.264	0.068	1.302	[1.140,1.488]	3.90	< 0.001
Serum Sodium	-0.044	0.023	0.957	[0.915,1.001]	-1.90	0.057
Hypertension (Yes vs No)	0.342	0.215	1.408	[0.924,2.146]	1.59	0.111

4.3 Validation of Model Assumptions (Grambsch-Therneau Test)

To prove the structural validity of the proportional hazards formulation, scaled Schoenfeld residuals ($r_j^*(t) \approx \beta_j(t) - \hat{\beta}_j$) were regressed against time:

Parameter Evaluated	ρ (Correlation)	χ2 Statistic	p-value (H0 :Constant Slope)
Age	0.012	0.014	0.906
Sex	−0.081	0.628	0.428
Ejection Fraction	−0.064	0.405	0.524
Serum Creatinine	−0.092	0.812	0.368
Serum Sodium	0.047	0.210	0.647
Hypertension	−0.031	0.093	0.760
GLOBAL SYSTEM	**NA**	2.481	0.871

conclusion: The global test statistic yields $p = 0.871$ ($p \gg 0.05$). The null hypothesis of constant hazard ratios over time is retained, verifying the proportional hazards assumption.

4.4 Discriminative Performance & Multicollinearity

- **Concordance Index (Harrell's C):** $C = 0.742 \pm 0.026$. The model correctly identifies the earlier-failing subject in approximately 74.2% of randomly selected clinical pairs.
- **Variance Inflation Factor (VIF):** All variables yielded VIF $\in [1.02, 1.24]$ (well below the conservative critical threshold of 5.0), confirming the absence of multicollinearity.

CHAPTER 5: SOFTWARE ARCHITECTURE & CLINICAL TRANSLATION

5.1 Reactive Computational Pipeline

The application was structured in R using the shiny reactive programming framework:

1. **Input Interface:** Sliders constrain numerical entry to physiologically plausible ranges (e.g., Creatinine: 0.5--9.0 mg/dL, Ejection Fraction: 14--80%).
2. **Reactive Execution:** Inputs assemble an instantaneous patient vector X_{new}, generating the prognostic index $\hat{\eta} = \hat{\boldsymbol{\beta}}^T X_{\text{new}}$ and mapping it through Breslow's baseline cumulative hazard step-function.
3. **Dynamic Visualization:** Uses ggplot2 to render $\hat{S}_{\text{new}}(t)$ bounded by continuous 95% confidence bands and reference markers at 90 and 180 days.

5.2 Counterfactual "What-If" Clinical Utility

The platform enables clinicians to test hypothetical treatment outcomes:

- *Case:* An admitted patient presents with Age $= 65$, EF $= 22\%$, and elevated Creatinine $= 2.6$ mg/dL (Predicted 90-Day Survival: $\approx 51\%$, High Risk).
- *Simulation:* Adjusting the slider to simulate targeted renal preservation (Creatinine $\rightarrow 1.2$ mg/dL) projects a survival recovery to $\approx 76\%$, providing quantitative decision support prior to clinical intervention.

CHAPTER 6: CRITICAL DISCUSSION, LIMITATIONS & CONCLUSION

6.1 Clinical & Pathophysiological Coherence

The regression outputs align with known cardiovascular physiology. The strong independent contribution of serum creatinine ($HR = 1.30, p < 0.001$) reflects **Cardiorenal Syndrome Type 1**, where primary cardiac failure induces renal hypoperfusion and accelerated systemic toxicity.

6.2 Study Limitations

1. **Sample Size:** $N = 299$ with 96 terminal events meets requirements for 6 covariates (> 15 events/variable), but warrants larger cohort validation.
2. **Static vs. Time-Varying Covariates:** The model utilizes baseline admission vitals. Real-world parameters fluctuate daily during hospitalization.
3. **External Generalizability:** Requires external validation across diverse healthcare registries before unrestricted clinical adoption.

6.3 Final Summary

This project bridges the gap between theoretical biostatistics and bedside clinical medicine. By combining semi-parametric hazard modeling, empirical assumption validation, and reactive cloud deployment, it delivers an open-access prognostic engine that improves upon legacy scorecards while maintaining regulatory-grade interpretability.

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